



Figure 10.1 This figure illustrates how the performance of a Kalman filter can be used to tune the values of F , H , Q , and R in order to obtain residual statistics that agree with theory. Alternatively, the Kalman gain K could be tuned directly.

the correct F , H , Q , and R , so the state estimate that comes out of that filter is probably the most correct. See [Kob03] for an application of this idea.

The analysis of this section can also be conducted for the continuous-time Kalman filter. The continuous-time innovations, $y(t) - H(t)\hat{x}(t)$, is a zero-mean white stochastic process with a covariance $R(t)$ (see Problem 10.2).

10.2 MULTIPLE-MODEL ESTIMATION

Suppose our system model is not known, or the system model changes depending on unknown factors. We can use multiple Kalman filters (one for each possible system model) and combine the state estimates to obtain a refined state estimate. Remember Bayes' rule from Section 2.1:

$$\Pr(x|y) = \frac{\Pr(y|x)\Pr(x)}{\Pr(y)} \quad (10.18)$$

Suppose that a random variable x can take one of N mutually exclusive values $x_1, \dots, x_N$. Then we can use Bayes' rule to write

$$\begin{aligned} \Pr(y) &= \Pr(y|x_1)\Pr(x_1) + \dots + \Pr(y|x_N)\Pr(x_N) \\ \Pr(x|y) &= \frac{\text{pdf}(y|x)\Pr(x)}{\sum_{i=1}^N \text{pdf}(y|x_i)\Pr(x_i)} \end{aligned} \quad (10.19)$$

where we have used the fact that the probability of an event occurring is directly proportional to the value of its pdf. Now suppose that we have the time-invariant system

$$\begin{aligned} x_k &= Fx_{k-1} + Gu_{k-1} + w_{k-1} \\ y_k &= Hx_k + v_k \\ w_k &\sim N(0, Q) \\ v_k &\sim N(0, R) \end{aligned} \quad (10.20)$$

The parameter set p is defined as the set (F, G, H, Q, R) . Suppose that p can take one of N possible values $p_1, \dots, p_N$. The question that we want to answer in this section is as follows: Given the measurements y_k , what is the probability that $p = p_j$? From Equation (10.19) this probability can be written as

$$\Pr(p_j|y_k) = \frac{\text{pdf}(y_k|p_j)\Pr(p_j)}{\sum_{i=1}^N \text{pdf}(y_k|p_i)\Pr(p_i)} \quad (10.21)$$

Now think about the probability that measurement y_k is observed given the fact that $p = p_j$. If $p = p_j$ then the state will take on some value x_k that is determined by the parameter set p_j . We therefore see that

$$\begin{aligned}\Pr(y_k|p_j) &= \Pr(y_k|x_k) \\ \text{pdf}(y_k|p_j) &= \text{pdf}(y_k|x_k)\end{aligned}\quad (10.22)$$

However, if our state estimate is accurate, then we know that $x_k \approx \hat{x}_k^-$. Therefore, the above equation can be written as

$$\text{pdf}(y_k|p_j) \approx \text{pdf}(y_k|\hat{x}_k^-) \quad (10.23)$$

The right side of the equation is the pdf of the measurement y_k given the fact that the state is $\hat{x}_k^-$. But since $y_k \approx H\hat{x}_k^- + v_k$, this pdf is approximately equal to the pdf of $(y_k - H\hat{x}_k^-)$. We therefore have

$$\begin{aligned}\text{pdf}(y_k|p_j) &\approx \text{pdf}(y_k - H\hat{x}_k^-) \\ &= \text{pdf}(r_k)\end{aligned}\quad (10.24)$$

where r_k is the residual defined in Section 10.1. From Section 10.1 we see that if w_k , v_k , and x_0 are Gaussian, then the residual r_k is a linear combination of Gaussian random variables. Recall from Section 2.4.2 that a linear combination of Gaussian random variables is itself Gaussian. In Section 10.1 we found the mean and variance of r_k . The pdf of r_k , which is approximated by the pdf of y_k given p_j , can therefore be approximated as

$$\text{pdf}(y_k|p_j) \approx \frac{\exp(-r_k^T S_k^{-1} r_k/2)}{(2\pi)^{q/2} |S_k|^{1/2}} \quad (10.25)$$

where $r_k = y_k - H_k \hat{x}_k^-$, $S_k = H_k P_k^- H_k^T + R_k$, and q is the number of measurements.

Now from Bayes' rule we can write the following equation for the probability that $p = p_j$ given the fact that the measurement y_{k-1} is observed.

$$\Pr(p_j|y_{k-1}) = \frac{\Pr(y_{k-1}|p_j)\Pr(p_j)}{\Pr(y_{k-1})} \quad (10.26)$$

If we are presently at time k , then the measurement at time $(k-1)$ is a given. The value of the measurement at time $(k-1)$ is a certain event with a probability equal to one. Therefore, $\Pr(y_{k-1}|p_j) = \Pr(y_{k-1}) = 1$ and the above equation becomes

$$\Pr(p_j|y_{k-1}) = \Pr(p_j) \quad (10.27)$$

Now in Equation (10.21) we can substitute this equation for $\Pr(p_j)$, and we substitute Equation (10.25) for $\text{pdf}(y_k|p_j)$. This gives a time-recursive equation for evaluating the probability that $p = p_j$ given the fact that the measurement was equal to y_k . The multiple-model estimator can be summarized as follows.

The multiple-model estimator

1. For $j = 1, \dots, N$, initialize the probabilities of each parameter set before any measurements are obtained. These probabilities are denoted as $\Pr(p_j|y_0)$ ($j = 1, \dots, N$).

2. At each time step k we perform the following steps.

- (a) Run N Kalman filters, one for each parameter set p_j ($j = 1, \dots, N$). The *a priori* state estimate and covariance of the j th filter are denoted as $\hat{x}_{kj}^-$ and P_{kj}^- .
- (b) After the measurement at time k is received, for each parameter set approximate the pdf of y_k given p_j as follows:

$$\text{pdf}(y_k|p_j) \approx \frac{\exp(-r_k^T S_k^{-1} r_k/2)}{(2\pi)^{q/2} |S_k|^{1/2}} \quad (10.28)$$

where $r_k = y_k - H_k \hat{x}_{kj}^-$, $S_k = H P_{kj}^- H^T + R_k$, and q is the number of measurements.

- (c) Estimate the probability that $p = p_j$ as follows.

$$\Pr(p_j|y_k) = \frac{\text{pdf}(y_k|p_j) \Pr(p_j|y_{k-1})}{\sum_{i=1}^N \text{pdf}(y_k|p_i) \Pr(p_i|y_{k-1})} \quad (10.29)$$

- (d) Now that each parameter set p_j has an associated probability, we can weight each $\hat{x}_{kj}^-$ and P_{kj}^- accordingly to obtain

$$\begin{aligned} \hat{x}_k^- &= \sum_{j=1}^N \Pr(p_j|y_k) \hat{x}_{kj}^- \\ P_k^- &= \sum_{j=1}^N \Pr(p_j|y_k) P_{kj}^- \end{aligned} \quad (10.30)$$

- (e) We can estimate the true parameter set in one of several ways, depending on our application. For example, we can use the parameter set with the highest conditional probability as our parameter estimate, or we can estimate the parameter set as a weighted average of the parameter sets:

$$\hat{p} = \begin{cases} \operatorname{argmax}_{p_j} \Pr(p_j|y_k) & \text{max-probability method} \\ \sum_{j=1}^N \Pr(p_j|y_k) p_j & \text{weighted-average method} \end{cases} \quad (10.31)$$

As time progresses, some of the $\Pr(p_j|y_k)$ terms will approach zero. Those p_j possibilities can then be eliminated and the number N can be reduced.

In Equation (10.31), the function $\operatorname{argmax}_x f(x)$ returns the value of x at which the maximum of $f(x)$ occurs. For example, $\max(1-x)^2 = 0$ because the maximum of $(1-x)^2$ is 0, but $\operatorname{argmax}_x (1-x)^2 = 1$ because $(1-x)^2$ attains its maximum value when $x = 1$. A similar definition holds for the function argmin .

■ EXAMPLE 10.1

In this example, we consider a second-order system identification problem [Ste94]. Suppose that we have a continuous-time system with discrete-time measurements described as follows:

$$\begin{aligned}
\dot{x} &= Ax + Bw_1 \\
&= \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} w_1 \\
y_k &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x_k + v_k \\
w_1 &\sim N(0, Q_c) \\
v_k &\sim N(0, R)
\end{aligned} \tag{10.32}$$

The damping ratio $\zeta = 0.1$, and the process and measurement noise covariances Q_c and R are respectively equal to 1000 and $10I$. The natural frequency $\omega_n = 2$, but this is not known to the engineer. The engineer knows that ω_n^2 is either 4, 4.4, or 4.8 with the following *a priori* probabilities:

$$\begin{aligned}
\Pr(\omega_n^2 = 4) &= 0.1 \\
\Pr(\omega_n^2 = 4.4) &= 0.6 \\
\Pr(\omega_n^2 = 4.8) &= 0.3
\end{aligned} \tag{10.33}$$

The state equation can be written as

$$\begin{aligned}
\dot{x} &= Ax + w \\
w &\sim N(0, BQ_cB^T)
\end{aligned} \tag{10.34}$$

We can discretize the system using the technique given in Section 1.4. If the measurements are obtained every 0.1 seconds, then we discretize the state equation with a sample time of $T = 0.1$ to obtain

$$\begin{aligned}
x_k &= Fx_{k-1} + \Lambda w'_{k-1} \\
y_k &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x_k + v_k \\
F &= \exp(AT) \\
\Lambda &= (F - I)F^{-1}
\end{aligned} \tag{10.35}$$

From Section 8.1 we know that the covariance Q' of the discrete-time noise w'_k is given as

$$Q' \approx BQ_cB^TT \tag{10.36}$$

This means that the discrete-time process dynamics can be written as

$$\begin{aligned}
x_k &= Fx_{k-1} + w_{k-1} \\
w_k &\sim N(0, Q) \\
Q &= (F - I)F^{-1}(BQ_cB^TT)F^{-T}(F^T - I)
\end{aligned} \tag{10.37}$$

The multiple-model estimator described in this section was run on this example. Three Kalman filters running in parallel each generate an estimate of the state. As the filters run, the probability of each parameter is updated by the multiple-model Kalman filter. Figure 10.2 shows the parameter probabilities for a typical simulation run. It is seen that even though the correct parameter has the lowest initial probability, the multiple-model filter estimate converges to the correct parameter after a few seconds.

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